

8 **D** Downward force exerted by the cylinder on the surface = pA

By Newton's 3rd law,

downward force exerted by cylinder on surface

= upward force exerted by the surface on the cylinder, N

Since the cylinder is in equilibrium.

$$N = W$$

$$pA = mg$$

$$p = \frac{mg}{A} = \frac{\rho A x g}{A} = \rho x g$$

$$\rho = \frac{p}{gx}$$

9 **B** Loss in GPE = gain in KE + energy dissipated by frictional forces